How does the horizontal displacement (in m) of the paper airplane depend on the surface area of its wings when it is launched from an inclined slingshot at a height of $1.016 \pm 0.0005$m?

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# 1 Introduction

The physics of a paper airplane has always fascinated me ever since I was a child. I remember making different airplane designs in grade 3 and testing out which plane flew further or maneuvered better through a set of hoops set up in my classroom. Now that I am older, I understand more about the forces acting on paper airplanes, as well as the importance of aerodynamics for any moving object in air. This prompted the idea to see if there is a relationship between the surface area of the wings of a paper airplane and its horizontal displacement. This resulted in the research question: How does the horizontal displacement (in m) of the paper airplane depend on the surface area of its wings when it is launched from an inclined slingshot at a height of $1.016 \pm 0.0005$m? The specific objective of this experiment is to determine a quantitative relationship between the wing surface area of a paper airplane and its horizontal displacement when launched under controlled environmental conditions (constant mass, launch height, initial thrust, wing geometry, center of mass, and environment). By only changing the surface area of the wings of the plane, this experiment aims to evaluate how changes in lift and drag affect flight distance. Since lift is proportional to surface area under constant air density and velocity, the investigation seeks to determine whether horizontal displacement increases linearly with surface area or whether deviations arise due to changes in drag, as a result of wing surface area increases. The hypothesis, solely based on theory, is that as the surface area of the wing surface area decreases, so will the horizontal displacement due to the influence of the wings on the aerodynamic forces of drag and lift.

## 1.1 Background Information

The physics behind this experiment is related to the forces at play on the airplane during flight. There is lift, the most important force in our experiment, which is a force that is perpendicular to the motion of the plane. When the plane is flyng straight forward, lift counteracts the force due to gravity and this is what allows the paper airplane to stay in the air longer and have a different path of descent compared to a projectile experiencing only the force due to gravity. The lift is generated by the movement of air over and under the wings, with the air flow being faster over the curved top surface of the wing. This difference in air flow speed creates low pressure above the wing and high pressure below the wing producing lift. (Federal Aviation Administration, 2016) The lift equation is

$$L = \frac{1}{2}\rho v^2 C_L A \tag{1}$$

where:

- L is lift,
- $\rho$ is the density of the air,

- $v$ is the airspeed of the aircraft,
- $C_L$ is coefficient of lift,
- A is the surface area of the wings.

In our case, this can be further simplified since the density of air, coefficient, and the initial velocity remained identical throughout our entire experiment. Therefore, $\rho$, $v$, $C_L$ can be defined as constant C, which is a constant that depends on the plane's aerodynamic properties, environment, and launch conditions. Therefore the simplified lift equation is

$$L = C \times A \tag{2}$$

The other key force that changes in our experiment due to the changing surface area is drag. As the surface area increases, the drag force also increases because the larger surface area encounters more air molecules, resulting in greater resistance to motion. Drag is the resistance force that opposes the motion of the airplane through the air. (NASA Glenn Research Center, 2023a) It can be split into two induced drag and parasite drag. Induced drag is the only one that is directly influenced by a change in lift, and occurs because the distribution of lift is not uniform on a wing, but varies from root to tip. These two aerodynamic forces differ from gravity or thrust, as a result of the slingshot, in the placement they are found on the airplane. While gravity and thrust are found on the center of mass (COM), lift and drag are found on the center of pressure (COP).

Thrust is the forward initial force that propels the paper airplane, generated by the slingshot in this experiment. (NASA Glenn Research Center, 2023b) The thrust depends on the tension in the slingshot and it provides the initial velocity that launches the plane into the air. Lastly, the force of gravity acts on the center of mass of the airplane and the weight of the airplane is directly proportional to its mass. The force due to gravity is a constant downward force that the airplane experiences.

# 2 Methodology

## 2.1 Variables

### 2.1.1 Independent Variable

The Independent Variable is the wing's surface area ($181\pm3\text{cm}^2$, $167\pm3\text{cm}^2$, $153\pm3\text{cm}^2$, $138\pm3\text{cm}^2$, $124\pm3\text{cm}^2$).

### 2.1.2 Dependent Variable

The Dependent Variable is the horizontal displacement of the paper airplane after being launched from our launcher.

### 2.1.3 Controlled Variables

Table 1: Controlled variables (excluding paper airplane design)

| **Controlled Variable** | **Why the Variable is Controlled** | **How the Variable is Controlled** |
|---|---|---|
| Mass of the paper airplane | The mass of the paper airplane was kept constant because any change would alter the weight $mg$, changing the force due to gravity and therefore influencing the $\Sigma F_y$ and $a_y$. Since horizontal displacement depends on the time in the air, a changing vertical acceleration would introduce systematic error. | Each airplane was measured using a triple-beam balance. Mass was maintained at $m = 4.90 \times 10^{-3} \pm 5.0 \times 10^{-5}\, kg$ by using the same paper and amount of duct tape. |
| Launch height | Launch height determines the initial vertical height and therefore affects the change in vertical position. Which directly influences the time of flight: $\Delta y = v_{y0}t + \frac{1}{2}a_y t^2$, and therefore the horizontal displacement. Which depends on time in the air ($\Delta x = v_x \Delta t$). | The launcher was fixed at $1.016 \pm 0.0005\, m$ using a table, wooden blocks, and a rail. The experiment slingshot was kept consistent throughout all trials. |

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| **Controlled Variable** | **Why the Variable is Controlled** | **How the Variable is Controlled** |
|---|---|---|
| Angle of launch | The launch angle affects the initial velocity components ($v_x$ and $v_y$), which changes the trajectory and flight time. If the angle varied, changes in displacement could be due to the different angles of launch rather than wing surface area. | The launch rail kept consistently fixed on the two wooden blocks and a protractor was used every morning to ensure that the launch angle remained identical. |
| Rubber band extension / Launcher | The initial energy input depends on the elastic potential energy stored in the band, $E_{elastic} = \frac{1}{2}kx^2$. Keeping the same band and extending it by $x$ every time, keeps the initial launch energy constant. | The same rubber band was used for all trials and was extended to $0.165\,m$ from its equilibrium length using the ruler on the side of the rail before each launch. |
| Environmental conditions | Air currents and changes in air density affect lift and drag, this could result in more random errors and variations within the data. | Trials were conducted indoors in a closed classroom with doors and windows closed to minimize drafts. |

The last controlled variable is the paper airplane's design. The plane's design remained identical to ensure that changes in the horizontal displacement was as a result of the change in surface area and not due to aerodynamic improvements. The wing's surface area was reduced by folding the back of the paper initially, keeping the shape of the fuselage the same, while taking off a strip of surface area from the back of the wing. The changes in surface area can be shown in the diagram below

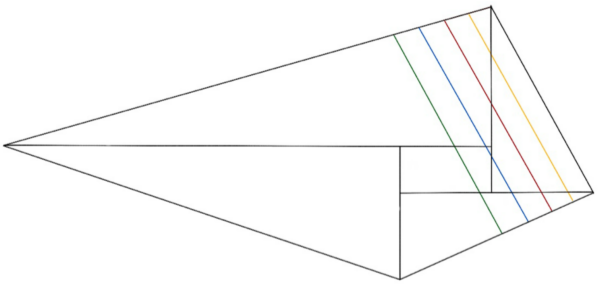


Figure 1: Depiction of the wing variations

| Change $n$ | Total area removed ($cm^2$) | Total Wing Surface Area $\pm 3$ ($cm^2$) |
|---|---|---|
| 1 | 0.0 | 181 |
| 2 | 14 | 167 |
| 3 | 28 | 153 |
| 4 | 43 | 138 |
| 5 | 57 | 124 |

## 2.2 Apparatus

The following apparatus was used in this experiment:

- Letter-size printer paper (0.216 m $\times$ 0.279 m)
- Duct tape
- Triple-beam balance ($\pm 5.0 \times 10^{-5}\, kg$)
- Metal rail
- Two wooden support blocks
- Elastic band
- Tape measure ($\pm 5.0 \times 10^{-4}\, m$)
- Three meter sticks ($\pm 5.0 \times 10^{-4}\, m$)

## 2.3 Procedure

1. A paper airplane was constructed using letter-size printer paper following the folding pattern shown below.

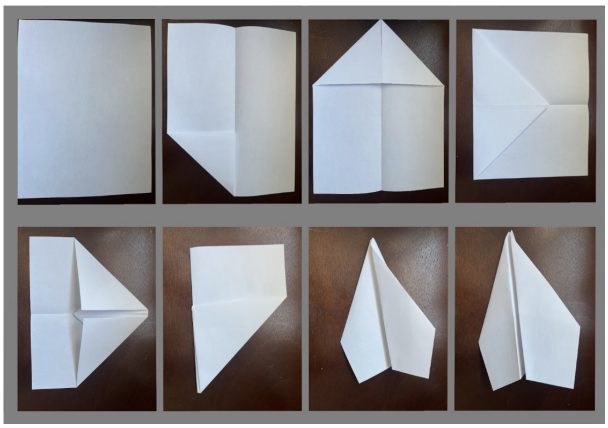


Figure 2: Plane Folding Template

2. Several small pieces of duct tape were applied to the rear fuselage of the airplane to prevent deformation, from the rubber band, during repeated launches.

3. The total mass of the airplane, including duct tape, was measured using a triple-beam balance and adjusted to $4.90 \times 10^{-3} \pm 5.0 \times 10^{-5}\, kg$.

4. The launch rail was placed on a stable table and two wooden blocks were positioned beneath the legs of the rail. The top of the rail was set to a fixed height of $1.016 \pm 0.0005\, m$ and safely positioned in a stable position.

5. An elastic band was attached to the front of the launch rail.

6. The airplane was placed into the groove of the rail and an elastic band was pulled back to a marked distance of $0.165\, m$ from its equilibrium position.

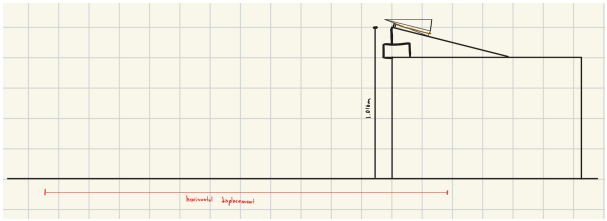


Figure 3: Experiment Setup

7. The airplane was released without applying additional force.

8. The horizontal displacement was measured using a tape measure from the vertical projection of the launch point on the ground to the back of the plane and the measurement uncertainty was estimated as $\pm 5.0 \times 10^{-4}\, m$.

9. Eight trials were conducted for each wing surface area.

10. After completing trials for one surface area, the wing surface area was reduced by folding the back of the paper along predetermined parallel lines shown in the design above, and the procedure was repeated.

### 2.4 Risk Assessment

Table 2: Risk assessment

| **Hazard** | **Risk Level** | **Safety** |
|---|---|---|
| Elastic band snapping during launch | Moderate | The elastic band was inspected before each trial for signs of weakness or tearing. |
| Damage to face or eyes by the paper airplane | Low | A clear flight path was established in the classroom with no classmates in the line of fire. The flight path was a designated clear area with no obstructions or bystanders. |
| Tripping hazard from equipment | Low | The equipment was arranged neatly within the flight path and kept close to the experiment area. |
| Paper Usage (environmental) | Low | Several sheets of paper were modified for the experiment due to stress from repeated launches. Separate planes were prepared for each surface area variation. However, all the used paper was recycled after completion of the experiment. |

## 3 Data Collection

### 3.1 Raw Data Table

After the experiment had been complete, the raw data we collected is shown below,

| **Wing Surface Area ($m^2$)** | **0.0181 ± 0.0003** | **0.0167 ± 0.0003** | **0.0153 ± 0.0003** | **0.0138 ± 0.0003** | **0.0124 ± 0.0003** |
|---|---|---|---|---|---|
| Trial 1 | 4.39 | 3.71 | 3.57 | 3.35 | 3.24 |
| Trial 2 | 4.22 | 3.55 | 3.43 | 3.17 | 3.11 |
| Trial 3 | 3.81 | 3.36 | 3.65 | 3.38 | 2.60 |
| Trial 4 | 3.77 | 3.39 | 3.39 | 3.32 | 3.05 |
| Trial 5 | 3.92 | 3.61 | 3.88 | 3.47 | 2.97 |
| Trial 6 | 4.27 | 3.29 | 3.55 | 3.62 | 3.29 |
| Trial 7 | 3.68 | 3.92 | 3.74 | 3.49 | 3.35 |
| Trial 8 | 4.08 | 3.78 | 3.32 | 3.66 | 2.74 |

Table 3: Horizontal displacement data (in m) for varying wing surface areas.

8 trials were performed due to the high variability involved in our lab. The inconsistent flight paths and horizontal displacement meant that we decided to perform more trials and find the overall mean for each surface area before performing any regression modeling to ensure more consistent information.

## 3.2 Trial Mean Data Table

From our 8 trials, the mean is taken and used because it enables us to eliminate some of the noise, as a result of minor deformations in the plane, air turbulence, and human error in the launching process. This reduces the impact of random error in our experiment, and the uncertainty as a result of this mean will also be evaluated. An example calculation for the surface area $0.0181 \pm 0.0003m^2$ will be shown, while the rest of the calculations were done and the values can be seen in the table below. The mean calculation for the wing surface area $0.0181 \pm 0.0003m^2$is:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Hence, for our first set of 8 trials, the calculation is:

$$\bar{x} = \frac{1}{8}(4.39)(4.22)(3.81)(3.77)(3.92)(4.27)(3.68)(4.08)$$

$$\bar{x} = 4.0175m$$

Rounded to the 3 sig. figs.:

$$\bar{x} = 4.02m$$

The standard deviation of the trials within a surface area should be calculated because it provides a measure of the spread of data about the mean and later helps determine the uncertainty in the mean

displacement. The sample standard deviation is calculated using the formula:

$$\sigma = \sqrt{\frac{1}{n-1}\sum(x_i - \bar{x})^2}.$$

Substituting the values from above into the standard deviation formula,

$$\sigma = \sqrt{\frac{1}{7}\Big[(4.39 - 4.0175)^2 + (4.22 - 4.0175)^2 + (3.81 - 4.0175)^2 + (3.77 - 4.0175)^2 + (3.92 - 4.0175)^2 + (4.27 - 4.0175)^2}$$

Evaluating the formula with the given values gives:

$$\sigma = \sqrt{\frac{0.4752}{7}} = \sqrt{0.06789} = 0.2605m.$$

Therefore, to an appropriate number of significant figures,

$$\sigma = 0.26m.$$

Lastly, the uncertainty in the mean displacement is needed to understand the precision of our estimate of the average, and this uncertainty will later be used for the error bars in our graph. The uncertainty as a result of this mean is:

$$\Delta\bar{x} = \frac{\sigma}{\sqrt{n}}$$

$$\Delta\bar{x} = \frac{0.26m}{\sqrt{8}}$$

$$\Delta\bar{x} = 0.0921m$$

Rounding to the proper number of sig. figs. for uncertainty

$$\Delta\bar{x} = 0.09m$$

Below is the table for all the other surface area mean displacement, standard deviation, and uncertainty.

| Surface Area ($m^2$) | Mean Displacement $\pm$ Uncertainty (m) | Standard Deviation (m) |
|---|---|---|
| $0.0181 \pm 0.0003$ | $4.02 \pm 0.09$ | 0.26 |
| $0.0167 \pm 0.0003$ | $3.58 \pm 0.08$ | 0.22 |
| $0.0153 \pm 0.0003$ | $3.57 \pm 0.07$ | 0.19 |
| $0.0138 \pm 0.0003$ | $3.43 \pm 0.06$ | 0.16 |
| $0.0124 \pm 0.0003$ | $3.04 \pm 0.09$ | 0.27 |

Table 4: Mean horizontal displacement, standard deviation, and uncertainty of the mean for different wing surface areas.

### 3.3 Graph

Using the values and uncertainty from the trial mean data table, a graph was created with horizontal and vertical error bars.

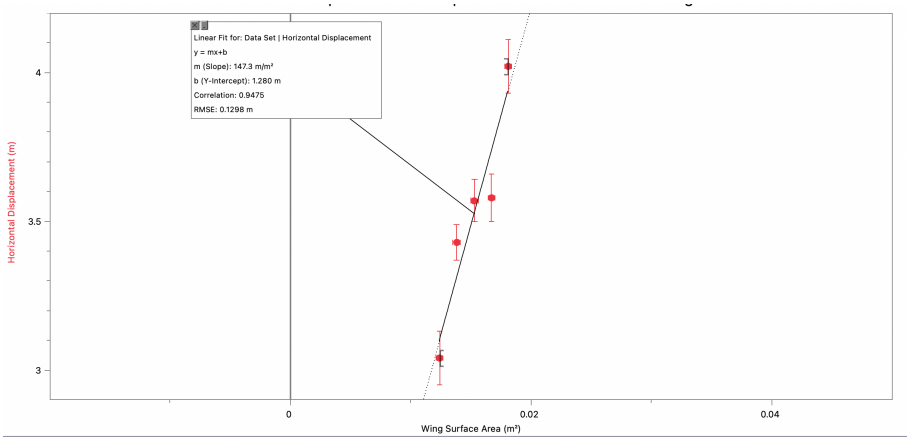



Figure 4: The horizontal disaplcement w.r.t. the surface area of the wings

## 4 Data Analysis

The mean horizontal displacement for each different surface area was plotted against the surface area of each wing. Afterwards, a linear relationship was spotted and a linear regression line was applied, giving:

$$y = mx + b$$

Where:

- m $= 147.3 \frac{m}{m^2}$,

- b = 1.280m,
- Correlation Coefficient $r = 0.9475$,
- RMSE = 0.1298m.

## 4.1 Theoretical Reasoning

The linear regression line indicates a strong positive relationship between our independent and dependent variables, since the correlation coefficient is 0.9475. To understand this trend, it is necessary to analyze the forces acting on the airplane during the flight and the extent to which lift and drag influence the airplane's movement through air. The motion of the airplane can be described using a free-body diagram, shown in Figure 5, which illustrates the fundamental forces acting on the plane about it's COM.

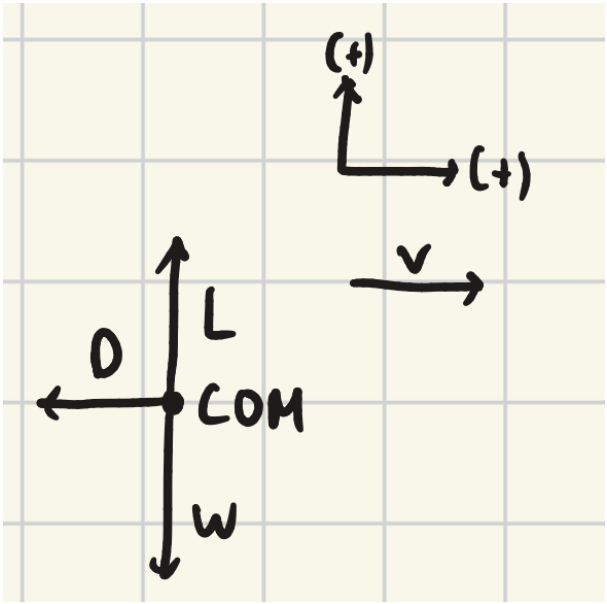



Figure 5: Free-Body Diagram (Translational Motion)

Lift actually acts perpendicular to the velocity, however, in our case the initial and final angle of the planes were so small that lift can be approximated upward. However, an important distinction is that the aerodynamic forces are actually acting on different aspects of the plane at different times. The two aerodynamic forces act on the COP, rather than the COM, which is the main reason why a paper airplane can have such erratic movements. The glide and direction of the plane's movement is determined by the position on the plane where the COP is located in relation to the COM, and the COP constantly moves about the paper airplane depending on the angle of it's nose.

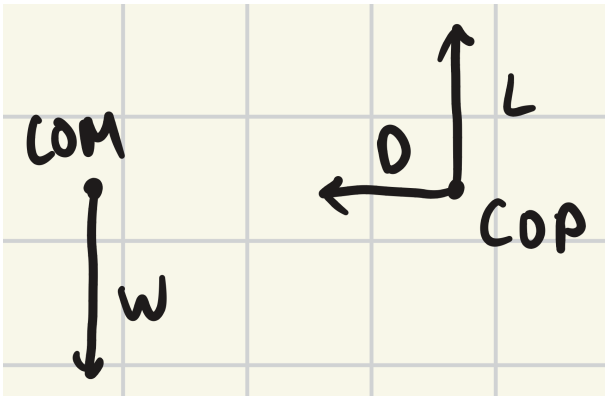



Figure 6: Extended Free-Body Diagram

Although the COM–COP separation affects pitching stability, the horizontal range is primarily determined by time of flight, which depends on the vertical acceleration. The strong linear relationship of the graph can be explained through dominant effect of lift on the airplane, compared to the drag effect. The simplified equation for lift is:

$$L = C \times A$$

As area increases, lift increases proportionally. In our experiment, the $\Sigma F_y$ statement is:

$$\Sigma F_y = L - w$$

Where:

- $\Sigma F_y$ is the sum of forces in the y-direction,
- L is the lift,
- w is the weight of the aircraft.

$$ma_y = L - mg$$

This can be further simplified and isolated for $a_y$

$$a_y = \frac{L - mg}{m}$$

$$a_y = \frac{CA - mg}{m}$$

Where:

- $a_y$ is the net acceleration in the y-direction,

- C is our simplified lift constant,
- m is the mass of the aircraft,
- g is the gravitational constant,
- A is the wing surface area.

In this equation, the relationship between surface area and the net acceleration in the y-direction can be clearly seen. Further analysis can be done using the kinematics equation:

$$\Delta y = v_{y0}t + \frac{1}{2}a_y t$$

Where $\Delta$y and $v_{y0}$ are kept constant. The time that the airplane is in flight increases, as the negative acceleration decreases. This supports the positive linear regression model in the graph, where as wing surface area increases so does the time the plane is in flight. In effect influencing the horizontal displacement $\Delta x = v_x t$. The other aerodynamic force that must be considered in our experiment is drag, and although drag depends on surface area, it doesn't impact our experiment to the same extent as lift. Since there are two types of drag, induced and parasite. In this case, as the induced drag decreases with increasing area, the parasite drag increases. These opposing trends partially offset each other, which means that the net drag change is moderate over the tested range. Therefore, the dominant effect influencing displacement is the increase in lift and flight time, which explains the strong linear correlation observed.

The RMSE value of 0.1298 $m$ represents the average deviation of the experimental data from the linear model. This corresponds to a relative deviation of approximately 3–4%, indicating that the linear model provides a strong but not perfect representation of the system. Several physical factors may contribute to this deviation. Firstly, small human errors in launch conditions, such as slight differences in rubber band extension, would alter the initial velocity and therefore affect both horizontal velocity and time of flight. Secondly, the separation between the COM and COP introduces torque during flight. Even minor asymmetries in folding or damage to the plane can produce rotational motion, which changes the angle of flight and therefore alters the lift and drag drastically. These aerodynamic and rotational effects are not accounted for in the simplified linear model, and therefore contribute to the observed residual scatter reflected in the RMSE value. Consequently, the RMSE supports the conclusion that the linear relationship is an approximation valid within the tested range. Overall, the results support the hypothesis that increasing wing surface area increases horizontal displacement under controlled launch and environment conditions.

# 5 Evaluation

## 5.1 Uncertainty

To conclude, the experiment slope does have some uncertainty and this can be accounted for using the max-min gradient method. The max–min gradient method estimates the uncertainty in the slope by determining the steepest and shallowest lines that remain consistent with the data's uncertainty bars. The difference between these limiting slopes reflects how much the gradient could vary due to experimental uncertainty.

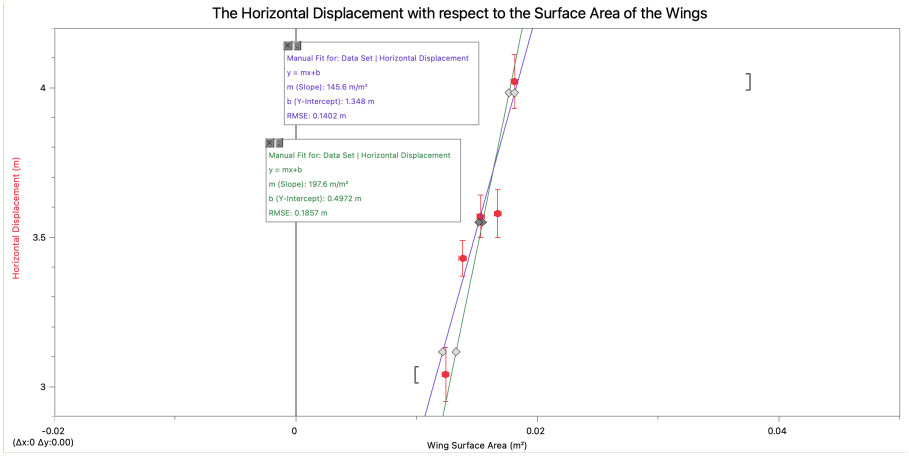



Figure 7: Max-min gradient

The max line is drawn as the steepest line that still passes within the uncertainty bars, while the min line is drawn as the shallowest line that still passes through the uncertainty bars. These two limiting lines represent the largest plausible range of gradients consistent with the experimental uncertainties. From the graph (with $x$ in metres and $A$ in $\mathrm{m}^2$), the limiting lines were:

$$m_{\max} = 197.6 \ \frac{\mathrm{m}}{\mathrm{m}^2}, \qquad m_{\min} = 145.6 \ \frac{\mathrm{m}}{\mathrm{m}^2}.$$

The uncertainty in the gradient was calculated as half the range:

$$\Delta m = \frac{m_{\max} - m_{\min}}{2} = \frac{197.6 - 145.6}{2} = \frac{52.0}{2} = 26.0 \ \frac{\mathrm{m}}{\mathrm{m}^2}.$$

Then the final gradient with uncertainty is:

$$m = 151 \pm 26 \ \frac{\mathrm{m}}{\mathrm{m}^2}.$$

### 5.2 Strengths

One major strength of this investigation was the control of variables affecting projectile motion. The launch height was fixed at $1.016 \pm 0.0005$ m and the launch angle remained constant, ensuring that variations in time of flight did not influence horizontal displacement. The elastic band was extended to a constant 0.165 m to consistently use the same elastic potential energy. Eight trials per surface area allowed for the mean to be calculated and used to reduce the impact of random errors, reducing random uncertainty. Conducting the experiment indoors further minimized environmental effects such as air currents, improving reliability.

### 5.3 Limitations

Despite these strengths, several limitations affected the investigation. Repeated launches caused minor structural deformation, shifting the COP relative to the COM and introducing torque, which altered the angle of flight and messed up the plane's trajectory. In some rare cases, this resulted in the planes hitting the wall. Moreover, human error during the plane-making process meant that any inconsistencies with folding would be very difficult to see during the launch of the plane, however it would be noticeable when the plane veered off course. Additionally, the simplification of the lift equation from $L = 12\rho v^2 C_L A$ to $L = CA$ assumes constant velocity, whereas drag reduces $v$ during flight. Finally, changing wing area altered the influence of induced and parasite drag to an immeasurable extent in our experiment.

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